SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

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CO5 - AT1 - LEVEL - 3

Level Exam | </> C | 03 Sep 2026, 10:00 AM

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Result Available
Your score: **65 / 100**

QUESTIONS

LEVEL 3 — HARD

Q1 function

Q2 functionsandpointers

2 / 2 saved

Level 3 — Hard **power (H)** — function

```
Write a function double power(double base, int exponent) that calculates the power of a given base raised to a given exponent.
        Sample Input:
Enter the base: 2
Enter the exponent: 3

Sample Output:
2 raised to the power 3 is 8
Number of consonants: 7
```

Your Code

```
#include <stdio.h>
#include <ctype.h>
double power(double base, int exponent) {
    double result = 1.0;
    int i;
    for (i = 0; i < exponent; i++) {
```

Program Input (stdin)

Enter input if required...

Output

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CO5 - AT2 - Debugging

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Result Available
Your score: **86 / 100**

QUESTIONS

LEVEL 1 — EASY

- Q1 Number
- Q2 Number
- Q3 Number
- Q4 Number

4 / 4 saved

Level 1 — Easy **Q57** — Number

```
<pre>
Debug the following code:
#include<stdio.h>
int main(){
    int i=0;
    do{
        printf("%d ", i);
    }while(i++ < 5);
    return 0;
}

</pre>
```

Your Code

```
#include<stdio.h>
int main(){
    int i=0;
```

Program Input (stdin)

Enter input if required...

CO5 - AT3 - HACKATHON

Q1 **CO5 - AT3 - HACKATHON / MINIPROJECT** 100 marks

CO5 – AT2_HACKATHON

1. A satellite continuously sends telemetry packets containing temperature, battery level, fuel status, altitude, and communication status to a ground station. Thousands of packets may arrive simultaneously from different processing channels. The system must store incoming records dynamically because the number of packets is unknown in advance.

Develop a C-based Real-Time Satellite Telemetry Processing System that:

- Uses dynamic memory allocation for telemetry records.
- Uses structures and pointers to manage satellite data.
- Uses functions to modularize packet processing.
- Uses POSIX threads to process multiple telemetry streams concurrently.
- Uses mutex/semaphore synchronization to prevent simultaneous updates to shared telemetry data.
- Stores processed telemetry in a file for later analysis.
- Uses command-line arguments to specify satellite ID and input/output files.
- Uses IPC or socket programming to simulate communication between the satellite simulator and ground station.

Hackathon Challenge: The system must continue processing telemetry even when multiple packets arrive at almost the same time without losing or corrupting records.

2. A hospital emergency department receives patient requests from multiple registration counters simultaneously. Each patient record contains patient ID, emergency level, symptoms, assigned doctor, and treatment status.